Docker-Airflow ETL Pipeline Documentation

1. Installation Steps

Step 1 — Install Docker Desktop

- Download Docker Desktop
- Docker provides containerization, allowing Airflow to run without manual dependency setup.
- Install and launch Docker

Verify installation:

- docker --version

Step 2 — Download Airflow Docker Setup

Create a project folder:

mkdir airflow-docker

cd airflow-docker

Download the Airflow docker-compose configuration:

curl -LfO https://airflow.apache.org/docs/apache-airflow/stable/docker-compose.yaml

Step 3 — Initialize Airflow

Run:

docker-compose up airflow-init

This initializes:

• Database

• User credentials

• Configuration files

Step 4 — Start Airflow Services

Run:

docker-compose up

This launches:

• Web server

• Scheduler

• Workers

• Database

Step 5 — Access Airflow UI

Open browser:

http://localhost:8080

Default login:

Username: airflow

Password: airflow

Airflow is now ready to execute pipelines.

Pipeline Flow:

Excel Dataset

↓

Extraction (read & store)

↓

Transformation (clean & compute)

↓

Load (final output storage)

Extraction Script

The ETL pipeline begins with extracting data from an Excel dataset.

Code Purpose:

• Reads Excel data

• Identifies correct headers

• Saves raw data into CSV format for processing

Extraction Function Code:

```
def extract():
    print("Extracting data...")
    df = pd.read_excel(DATA_PATH, header=7)
    print("Columns:", df.columns)
    df.to_csv("/tmp/extracted.csv", index=False)
```

Step 1 — Function Start

Indicates extraction has begun. This message appears in Airflow logs.

Step 2 — Read Excel File

Loads Excel dataset.

Header row is set to row 8.

Ensures column names are correctly interpreted.

Step 3 — Display Columns

Verifies dataset structure and confirms successful extraction.

Step 4 — Save Extracted Data

Converts Excel to CSV.

Stores temporary file.

Prepares data for transformation.

Transformation Function Code

The transformation stage cleans extracted data and performs calculations.

```
def transform():
    print("Transforming data...")
    df = pd.read_csv("/tmp/extracted.csv")
    df.columns = df.columns.str.strip()
    df["Calculated Total"] = (
```

```
        df["Retail Price (USD)"] *
        df["Order Quantity"] +
        df["Tax (USD)"]
    )
    df.to_csv("/tmp/transformed.csv", index=False)
```

Step 1 — Function Start

Indicates transformation has begun.

Step 2 — Read Extracted Data

Loads extracted CSV for processing.

Step 3 — Clean Column Names

Removes unwanted spaces to prevent errors.

Step 4 — Business Calculation

Computes:

Retail Price × Quantity + Tax

Step 5 — Save Transformed Data

Stores cleaned data for loading stage.

Load Function Code

Stores final dataset and previews results.

```
def load():
    print("Loading data...")
    df = pd.read_csv("/tmp/transformed.csv")
    print(df.head())
    df.to_csv("/opt/airflow/data/final_sales.csv", index=False)
```

Step 1 — Function Start

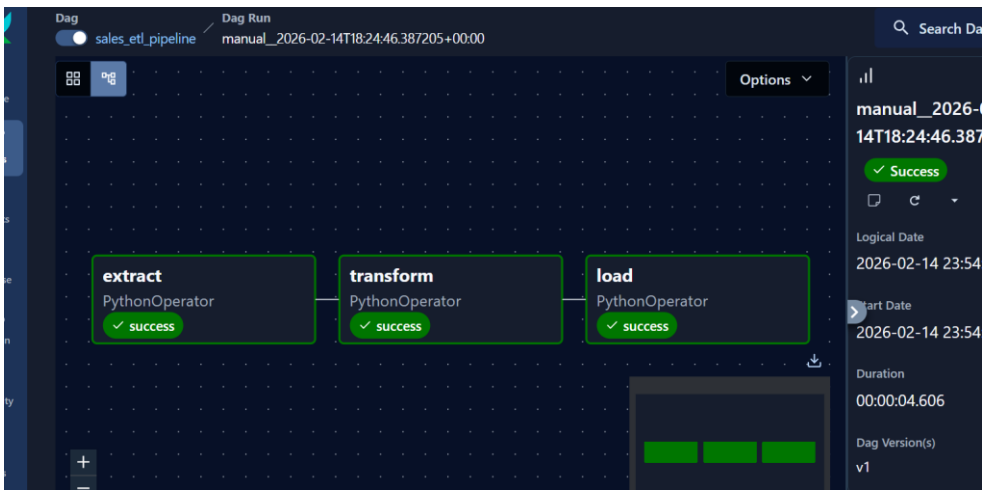
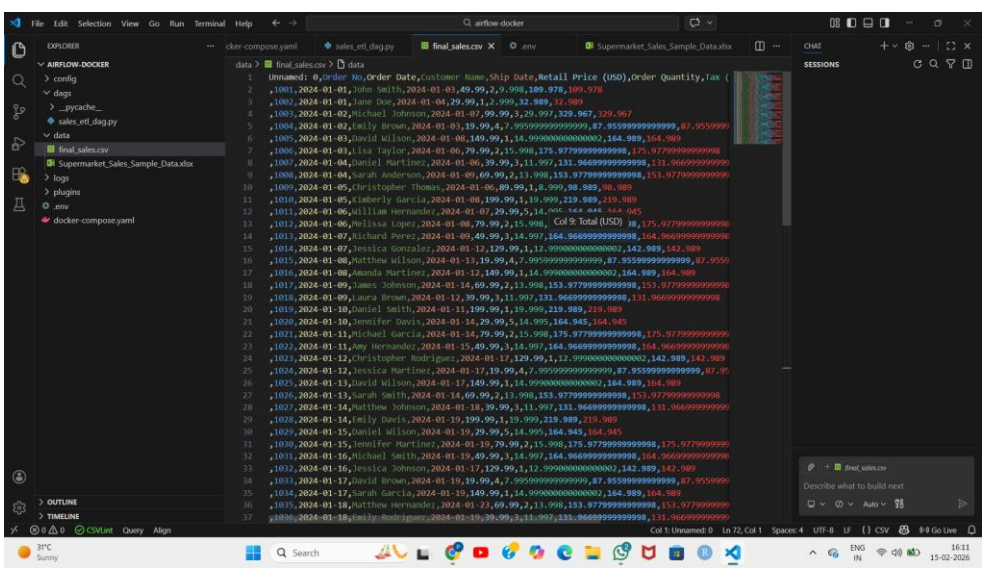
Confirms load phase execution.

Step 2 — Read Transformed Data

Loads processed dataset.

Step 3 — Preview Data

Displays first rows for verification.

Step 4 — Save Final Output

Stores final dataset permanently.